

Panel MS-AR(1): common intercept and linear trend

1985 through the end of the sample, excluding the US, Germany, Japan, and China

1 Model

Seasonally adjusted real GDP per worker, 2026-09-17 vintage, 1985-Q1 through 2026-Q2. United States, Germany, and Japan are excluded. China is not in this vintage.

$$y_{it} = a + g t + z_{it}, \tag{1}$$

$$s_{i,t+1} \sim \Pi(\cdot \mid s_{it}), \tag{2}$$

$$z_{i,t+1} = (1 - \rho(s_{i,t+1}))\mu(s_{i,t+1}) + \rho(s_{i,t+1})z_{it} + \sigma(s_{i,t+1})\varepsilon_{it}. \tag{3}$$

The regime s_{it} is the one that produced z_{it} . ε_{it} is the innovation. g is the within-country slope, so a country that enters later does not shift g through its level. a is then set so that $z_{it} = y_{it} - a - gt$ has mean zero. Three regimes. The MS-AR imposes $E[z] = 0$. σ and ρ switch, with $|\rho| < 0.99$. Standard errors are delta-method from a numerical Hessian of the cycle parameters. a and g have none.

2 Common intercept and linear trend, 1985 through the end of the sample, ex US, Germany, Japan, China

2026-09-17 SA panel, 1985-Q1 through 2026-Q2, excluding the US, Germany, and Japan. $y_{it} = a + gt + z_{it}$, g the within-country slope, $E[z] = 0$. 71 countries, 7893 observations, 1985-Q1–2026-Q2. Fitted countries: 71. Observations: 7893. Log-likelihood: 20160.06. The within-country slope is $g = 0.0164$ and the intercept that zeros the pooled mean of z is $a = -5.3569$ (no standard errors). a is the intercept at the first date in the estimation sample. $z_{it} = y_{it} - a - gt$ is the cycle in the figure. Its average within-country slope is zero. The MS-AR is the law of z , with $E[z] = 0$.

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.7737 (0.0749)	-0.7471 (0.0374)	1.1467 (0.0140)
σ	0.0733 (0.0021)	0.0159 (0.0003)	0.0098 (0.0002)
ρ	0.9900 (0.0000)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.7223 (0.0255)	0.2000 (0.0221)	0.0777 (0.0103)
Π_1	0.0610 (0.0059)	0.9385 (0.0058)	0.0005 (0.0012)
Π_2	0.0284 (0.0033)	0.0000 (0.0004)	0.9716 (0.0033)
π	0.1419 (0.0107)	0.4616 (0.0154)	0.3965 (0.0110)

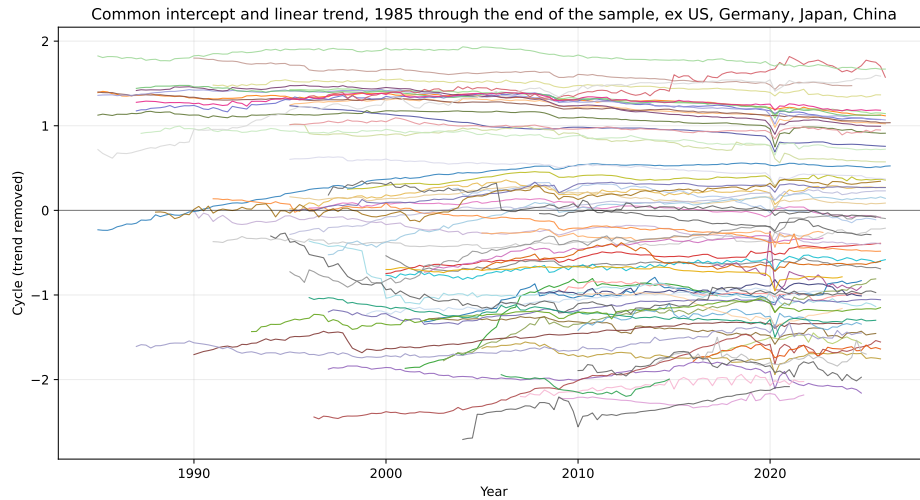


Figure 1: Country cycles $z_{it} = y_{it} - a - gt$.